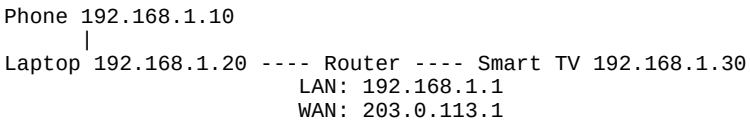
Routing and Networking Assignment Report

1. Network Diagram



2. Manual IP Routing Simulation

Laptop 192.168.1.20 sends a packet to Friend PC 192.168.2.50. Laptop checks its local subnet and sees destination is remote, so it forwards the packet to default gateway 192.168.1.1. Router R1 checks routing table and finds network 192.168.2.0/24 reachable through next hop 10.0.0.2. Router R2 receives the packet and delivers it to 192.168.2.50.

```
Laptop Routing Table
192.168.1.0/24  Local
0.0.0.0/0      192.168.1.1

R1 Routing Table
192.168.1.0/24 connected
192.168.2.0/24 via 10.0.0.2
```

3. Zomato-style Delivery Routing Table

City	Network	Mask	Next Hop	Interface
Ahmedabad	10.1.0.0	255.255.255.0	Direct	G0/0
Mumbai	10.2.0.0	255.255.255.0	172.16.0.2	G0/1
Delhi	10.3.0.0	255.255.255.0	172.16.0.3	G0/2

4. Static vs Dynamic Routing

Static Routing Pros: simple, predictable, low overhead. Cons: manual updates, poor scalability, difficult failover. Dynamic Routing (RIP/OSPF) Pros: automatic updates, scalability, automatic rerouting. Cons: more complexity, protocol overhead, troubleshooting difficulty. A food delivery network benefits from dynamic routing because traffic can automatically follow alternate paths during outages.

5. Sample OSPF Configuration and Explanation

```
Router1
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0

Router2
router ospf 1
 network 10.0.0.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0

Explanation:
router ospf 1 -> starts OSPF process 1
network ... area 0 -> advertises connected networks into backbone area 0.
```